

# Predicting Graduate Admission Likelihood: A Neural-Network Classification Study of Applicant Profiles

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**Abstract**—Graduate admissions committees traditionally screen applicants with ad hoc GPA or test-score cutoffs, or review every file with equal depth regardless of how clearly borderline it is. This paper asks whether a multilayer perceptron (MLP) can predict, from an applicant's academic and research profile, whether they are likely to be admitted, and which architecture classifies most reliably, on a dataset of 500 graduate applicants described by GRE, TOEFL, and CGPA scores, statement-of-purpose (SOP) and letter-of-recommendation (LOR) strength, university rating, and research experience. We binarize a continuous, dataset-provided admission-probability score at 0.8 to obtain a supervised binary label (155 of 500 applicants “likely admitted”). Following a four-stage pipeline—business understanding, data engineering, statistics and machine learning, and deployment—we compare three scikit-learn `MLPClassifier` architectures on an 80/20 stratified split: a single hidden layer of 3 units with ReLU activation, the same layer with tanh activation, and a deeper two-layer network (8, 4). The deeper network achieves the highest test accuracy (0.920) and the smallest train–test gap (+0.015), and is also the most stable under 5-fold stratified cross-validation (mean 0.9225, standard deviation 0.0215). CGPA, TOEFL score, and GRE score are the features most correlated with admission ( $r = 0.742, 0.699, 0.684$ ). The selected model is deployed in a Streamlit application that surfaces a single production model's prediction alongside the applicant's likelihood relative to the historical base rate, positioned as a screening aid rather than an admissions decision.

**Keywords**—graduate admission prediction, multilayer perceptron, neural network classification, feature scaling, Streamlit.

## I. Introduction

The admission to a graduate program is a routine but consequential task for any academic department. Committees commonly rely on a fixed GPA or test-score cutoff, or read every application with the same depth of scrutiny regardless of how clearly it falls above or below the bar. Such coarse approaches ignore how test scores, GPA, and qualitative factors such as SOP/ LOR strength and research experience

interact, and risk inconsistent screening decisions under time pressure [1].

The central business question of this study is therefore: can a graduate applicant's admission outcome be predicted from their academic and research profile, well enough to flag likely-admitted applicants for expedited review while routing borderline cases to a human reader? Unlike an unsupervised segmentation problem, a ground-truth label exists here — the dataset's own estimated admission

probability — so the answer can be validated with held-out test accuracy rather than an internal cohesion metric alone. As in related pipelines, the design follows the CRoss-Industry Standard Process for Data Mining (CRISP-DM) tradition of separating business understanding from modeling and deployment [2].

## II. Methodology

### A. Pipeline Architecture

The system reuses a four-stage architecture applied elsewhere in this project family, redesigned for a supervised binary-classification setting with a continuous raw target. Stage 1 strips a byte-order mark and stray whitespace from column names, binarizes `Admit_Chance` at a 0.8 threshold to obtain the label, and drops the identifier column; because the dataset contains no missing values, its imputation step is a defensive no-op. Stage 2 owns feature engineering: `University_Rating` and `Research` are numerically coded but semantically categorical, so both are one-hot encoded; the data is then split 80/20 with stratification on the binarized target (`random_state = 123`), and a `MinMaxScaler` is fit on the training split only and used to transform both splits — gradient-trained MLPs converge poorly on unscaled inputs, since `GRE_Score` (290–340) and `SOP` (1–5) differ by two orders of magnitude in raw range. Stage 3 trains three independent `MLPClassifier` architectures in isolated functions, so a future architecture swap touches only one function, and compares them by held-out test accuracy. Stage 4 persists the test-set predictions, all three fitted models and the scaler, and generates one business insight from the

comparison table plus the correlation-with-target ranking computed in Stage 2's EDA — MLPs expose no coefficient array the way logistic regression does, so the “which factors matter” question is answered from the data profile instead.

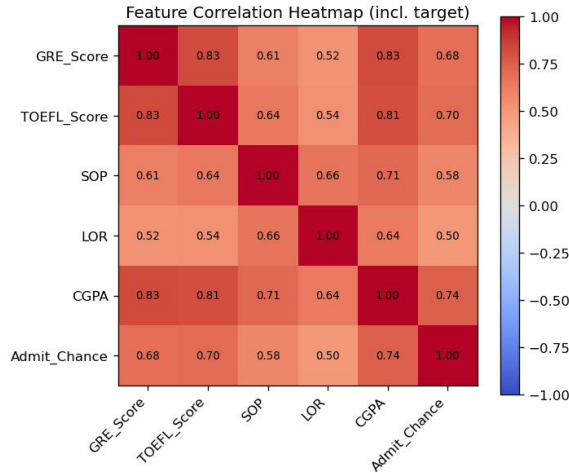
### B. Models and Evaluation

Three `MLPClassifier` architectures [3], [4], implemented in `scikit-learn` [6] with the default Adam solver [5], `batch_size = 50`, and `random_state = 123`, are compared: a single hidden layer of 3 units with ReLU activation (default), the same layer with tanh activation, and a deeper network with two hidden layers of 8 and 4 units. Because a ground-truth label exists, held-out test accuracy is the primary selection criterion, with the confusion matrix and per-class precision/recall logged for the selected model. Each architecture is additionally evaluated with 5-fold stratified cross-validation on the training split only, reported as a stability check that confirms the single 80/20 split was not a lucky draw.

## III. Data Processing

The cleaned dataset contains 500 graduate applicants and 8 columns after the identifier is dropped: `GRE score`, `TOEFL score`, `university rating` (1–5), `SOP` and `LOR strength` (1–5), `CGPA` (out of 10), `research experience` (0/1), and the binarized target. `Admission.csv` contains no missing values, so Stage 1's imputation step is a no-op confirmation rather than an active repair. Applicants span `GRE scores` of 290–340 (mean 316.5), `TOEFL scores` of 92–120 (mean 107.2), `SOP` and `LOR strength` of 1–5 (means 3.37 and 3.48), and `CGPA` of 6.80–

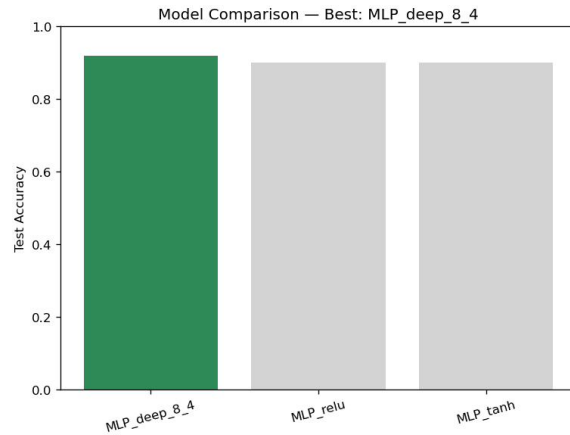
9.92 (mean 8.58). Binarizing Admit\_Chance at 0.8 yields 155 “likely admitted” applicants (31.0%) and 345 “unlikely” applicants (69.0%).



**Fig. 1.** Feature correlation heatmap, including the binarized target.

Correlation analysis (Fig. 1) shows CGPA is the feature most associated with admission ( $r = 0.742$ ), followed by TOEFL score ( $r = 0.699$ ) and GRE score ( $r = 0.684$ ), with SOP ( $r = 0.581$ ) and LOR ( $r = 0.500$ ) somewhat weaker. The three numeric test/GPA measures are themselves highly intercorrelated ( $r \approx 0.81$ – $0.83$ ), consistent with all three reflecting a shared underlying academic-ability signal, but none is redundant enough to drop — the MLP is left to combine all six inputs nonlinearly. Because gradient-trained networks are sensitive to feature scale, all numeric inputs are standardized to  $[0, 1]$  with MinMaxScaler fit on the training split before fitting, so GRE\_Score's larger numeric range does not dominate SOP/LOR purely because of scale.

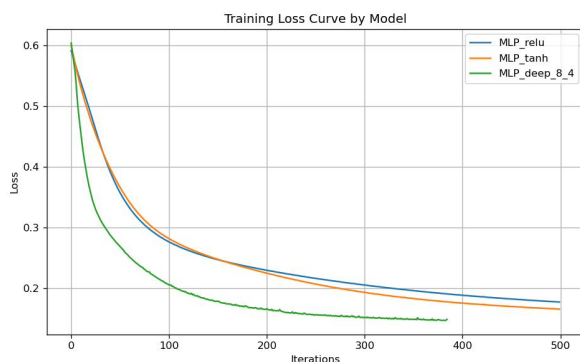
#### IV. Results



**Fig. 2.** Test accuracy by architecture, MLP\_deep\_8\_4 highlighted.

MLP\_deep\_8\_4 attains the highest test accuracy (0.920) despite having the most parameters, and, notably, the smallest train–test gap (+0.015) of the three — smaller than MLP\_relu's gap (+0.0225) and well below MLP\_tanh's (+0.040, Fig. 2).

The 5-fold stratified cross-validation reinforces this choice: MLP\_deep\_8\_4's mean CV accuracy (0.9225, std 0.0215) is both the highest and tied for the most stable, against MLP\_relu (0.9125, std 0.0274) and MLP\_tanh (0.9150, std 0.0215). The training loss curves (Fig. 3) show MLP\_deep\_8\_4 also converges fastest and to the lowest loss, consistent with its extra capacity being used productively here rather than overfitting the small training set.



**Fig. 3.** Training loss per iteration by architecture.

On the 100-applicant held-out test set, MLP\_deep\_8\_4 produces the confusion matrix in Fig. 4: 65 true negatives, 4 false positives, 4 false negatives, and 27 true positives. This corresponds to precision/recall of 0.94/0.94 on the majority “unlikely” class and 0.87/0.87 on the minority “likely admitted” class (overall accuracy 0.92, macro-averaged F1 0.91). Precision and recall are balanced on both classes, so the model is not simply defaulting to the majority class to inflate accuracy.

For deployment, MLP\_deep\_8\_4 is served in a four-tab Streamlit application [7]. The Business Intelligence tab shows only this single production model's prediction for a manually entered applicant profile — the predicted class, its probability, and that probability against the dataset's historical base rate (31.0% likely-admitted) — rather than surfacing all three architectures, since the multi-architecture comparison is a model-selection question answered separately in the Model Selection tab. This separation keeps the business-facing view focused on “what should we do for this applicant” rather than “which model is best.”

## V. Limitations

Four caveats qualify these results. First, Admit\_Chance is the source dataset's own estimated admission probability rather than a confirmed institutional decision, and the 0.8 threshold used to define “likely admitted” is a modeling choice, not an official cutoff — a different threshold would relabel some applicants and could shift which architecture wins. Second, correlation is not the mechanism the network actually uses: CGPA, TOEFL, and GRE correlate most strongly with the target individually, but the MLP combines all six inputs nonlinearly, so the correlation ranking is a simplified, not literal, account of feature importance. Third, only 100 applicants are held out for testing, and just 31 of them are labeled “likely admitted”; per-class metrics on this minority class, such as the 0.87 recall reported above, carry more sampling uncertainty than the overall accuracy figure suggests. Fourth, no causal claim is supported: a high predicted probability reflects historical association with Admit\_Chance, not evidence that raising a specific score would cause admission, since confounding factors such as recommendation-letter strength or interview performance are not present in the data.

## VI. Conclusion

On 500 graduate applicants, a deeper two-layer MLPClassifier (8, 4 units) outperforms two shallower single-layer alternatives (ReLU and tanh, both 3 units), achieving 0.920 test accuracy with the smallest train-test gap and the most stable cross-validated score of the three. CGPA, TOEFL, and GRE emerge as

the strongest correlates of admission, motivating a deployment that pairs each applicant's predicted likelihood with these top factors rather than returning a bare accept/reject call. Because the target is a proxy probability rather than a confirmed admissions outcome, and because the held-out test set is modest and imbalanced, the model is deployed as a screening aid that routes borderline predictions to human review rather than a final decision-maker. Future work should validate against real admissions-committee outcomes where available, audit predictions for disparate impact across demographic groups not present in this dataset, and periodically recalibrate the 0.8 threshold as applicant pools shift.

## VII. References

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